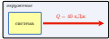


Isotermal expansion, $T = 300\text{ K}$

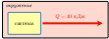


$$\Delta S_{\text{sur}} = 133\text{ J/K}$$

$$\Delta S_{\text{sur}} = Q/T = 40\,000\text{ J} / 300\text{ K}$$

The gas expands and does work on the surroundings.

Isotermal expansion, $T = 600\text{ K}$



$$\Delta S_{\text{sur}} = 67\text{ J/K}$$

$$\Delta S_{\text{sur}} = Q/T = 40\,000\text{ J} / 600\text{ K}$$

The gas expands and does work on the surroundings.